

Handover Optimization for LEO Satellite Networks Using Online Decision Transformer

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Abstract—Low Earth Orbit (LEO) satellite networks enable global connectivity but face significant challenges due to frequent handovers driven by rapid satellite motion and dynamic beam coverage. This paper proposes a novel handover management framework based on an Online Decision Transformer (ODT) for LEO satellite networks. By formulating handover control as a sequence modeling problem, the proposed approach learns trajectory-aware policies that account for long-term communication performance rather than instantaneous link conditions. The ODT is trained using a combination of offline trajectories generated by heuristic and reinforcement learning baselines and is further refined through online interaction with the environment. The proposed method is evaluated using realistic aircraft trajectories and heterogeneous passenger network traffic, including streaming, VoIP, web browsing, file transfer, email, and video conferencing applications. Simulation results show that the ODT-based approach achieves higher bandwidth allocation efficiency while reducing latency and service interruption duration by up to 7.84% and 62%, respectively, compared to an SNR-based baseline. These results demonstrate the effectiveness of sequence-aware, online reinforcement learning for robust handover management in LEO satellite networks.

Index Terms—LEO satellite networks, handover management, deep reinforcement learning, transformer networks, quality of experience

I. INTRODUCTION

Low Earth orbit (LEO) satellite networks are increasingly recognized as a viable infrastructure for global broadband connectivity, with large-scale constellations such as OneWeb and Starlink currently deploying hundreds to thousands of satellites to serve aeronautical, maritime, and remote terrestrial users [1]. The orbital altitude of LEO satellites yields propagation delays substantially lower than those of geostationary systems. However, the resulting orbital velocity imposes severe constraints on link continuity, necessitating frequent beam or satellite handovers throughout the duration of a user session.

In aeronautical satellite communication systems, handovers may occur at intervals as short as 1 to 2 minutes [2], placing stringent requirements on handover latency, stability, and service continuity. In practice, LEO handovers are typically triggered based on threshold-driven link-quality measurements and visibility constraints, where the user terminal (UT) switches beams when signal metrics indicate degradation or improved connectivity [3], [4]. Received signal quality is an unreliable proxy for end-to-end service performance in LEO systems as the beam offering the highest instantaneous signal-

to-noise ratio (SNR) may concurrently exhibit high congestion, elevated queuing delay, and reduced available throughput.

Deep reinforcement learning (DRL) has been proposed as a principled framework for adaptive handover management [5]–[7], formulating beam selection as a sequential decision-making problem in which an agent learns a policy over observed network states. Nevertheless, prevailing DRL-based approaches are typically optimized with respect to short-horizon discounted rewards and may not explicitly capture long-range temporal dependencies within handover sequences. As a result, modeling delayed congestion effects and long-term stability trade-offs can be challenging, particularly under heterogeneous and time-varying traffic demand.

To address these limitations, this paper proposes a state-of-the-art Online Decision Transformer (ODT) [8] based handover management framework for LEO satellite networks. By casting beam selection as a return-conditioned sequence prediction problem, the proposed approach enables the agent to incorporate extended trajectory context and align decisions with long-term performance objectives. The main contributions of this work are as follows:

- We formulate the handover optimization problem in a high-mobility aeronautical setting, where rapidly changing satellite visibility, dynamic congestion, and passenger demand create complex trade-offs between link quality, latency, and service continuity.
- We extend the open-source multi-agent LEO routing simulator of [9] to support handover decision-making by incorporating spot beam structures, an airborne user terminal, passenger-level traffic demand, and application-specific Quality of Service (QoS) requirements.
- We propose an ODT-based handover policy that conditions beam selection on a return-to-go target, enabling trajectory-aware decisions that account for long-range dependencies across the handover sequence.
- We conduct a comprehensive evaluation against an SNR-based greedy baseline, a Deep Q-Network (DQN) agent, and a Proximal Policy Optimization (PPO) agent using realistic aeronautical trajectories and heterogeneous passenger traffic, demonstrating reductions in average latency and service interruption duration of up to 7.84% and 62%, respectively, relative to the baseline.

The remainder of this paper is organized as follows. In

Section II, we review related work. In Section III, we introduce the system model. In Section IV, we formulate the problem and present the proposed ODT framework. In Section V, we present and analyze the simulation results. Finally, in Section VI, we conclude the paper.

II. RELATED WORK

A. Traditional Handover Mechanisms

Handover management in LEO satellite networks has been addressed using deterministic, heuristic, and optimization-based approaches. Conventional schemes rely on received signal strength (RSS) measurements combined with hysteresis margins and time-to-trigger mechanisms to mitigate ping-pong effects [10]. While effective in relatively static terrestrial networks, these methods become suboptimal in LEO environments, where rapid satellite motion and frequent topology changes reduce the relevance of delayed measurement reports.

Optimization-based formulations model beam-user association as a global network flow problem and compute handover schedules that minimize aggregate switching cost [11]. Although such approaches can yield near-optimal solutions, they require centralized computation and full network state information, limiting their practicality for distributed, real-time decision-making in large-scale constellations.

On the other hand, multi-attribute decision-making (MADM) techniques combine heterogeneous metrics, such as signal quality, elevation angle, and beam load, into a single handover score [12]. However, these methods depend on manually tuned weight parameters and often lack robustness under non-stationary traffic and congestion dynamics.

B. Deep Reinforcement Learning for Handovers

DRL-based handover frameworks have demonstrated substantial improvements over heuristic baselines by learning adaptive policies through interaction with the network environment. [5] proposed a DRL-based handover protocol that replaces periodic measurement reporting with learned network-condition prediction using the Importance Weighted Actor-Learner Architecture (IMPALA) algorithm, achieving up to a $6.86\times$ reduction in access delay. Additionally, [6] formulated load-aware handover as a multi-agent reinforcement learning problem, although the approach assumes access to global information regarding all users' actions. Finally, [13] introduced a distributed successive deep Q-learning framework that enables handover decisions based solely on local observations. Despite these advances, most existing DRL approaches employ value-based or actor-critic architectures that primarily optimize short-horizon objectives. Such formulations may struggle to capture long-range temporal dependencies inherent in handover sequences, particularly when beam selection decisions produce delayed congestion, stability, or latency effects. This limitation motivates the exploration of sequence-based learning frameworks capable of modeling extended temporal context in dynamic LEO environments.

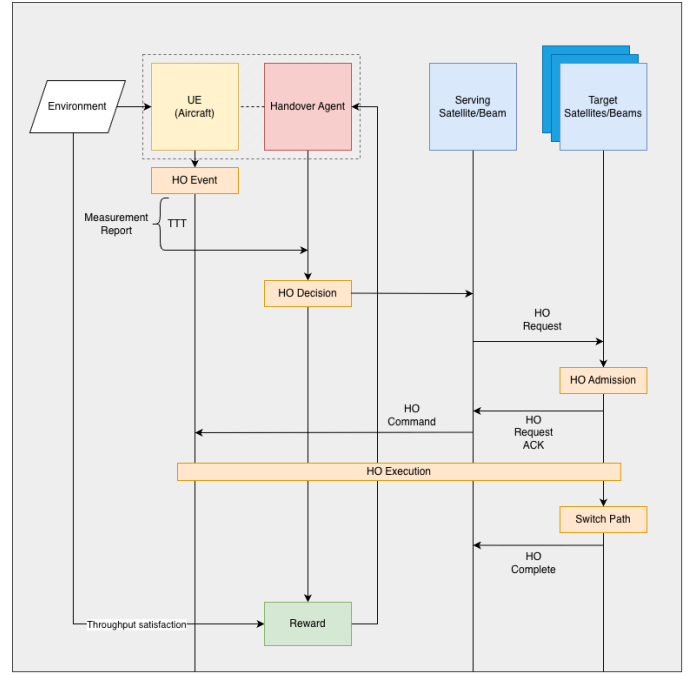


Fig. 1. System-level diagram of the RL-based LEO handover management process

III. SYSTEM MODEL

In this section we describe the system model and assumptions used to evaluate handover optimization in a LEO satellite communication network. We model an airborne UT connected to a LEO satellite constellation consisting of multiple spot beams per satellite. Due to the motion of the satellites and the aircraft, the visibility of beams varies over time. At each discrete time step, the UT observes link-quality metrics. A handover controller selects a serving beam subject to dwell time and time-to-trigger constraints. The objective is to learn a policy that maximizes long-term quality-of-experience metrics, including throughput allocation and latency satisfaction, while minimizing service interruption duration.

We model the handover management system as the learning agent, as shown in Figure 1. The aircraft and its onboard user terminal interact with the environment by observing network conditions and generating measurement reports, while the agent makes handover decisions based on this information. The observations consist of the aircraft's location, link quality metrics and beam visibility. Actions correspond to selecting one of the currently visible beams or remaining connected to the current beam. The reward is designed to encourage high throughput and low latency, while minimizing service interruptions.

We consider a LEO constellation, consisting of 648 satellites, each equipped with 16 spot beams. Each beam is characterized by:

- Maximum bandwidth: 250 MHz
- Maximum downlink speed: 150 Mbps
- Transmit power: 40 dBm

- Carrier frequency: 15 GHz

The aircraft user terminal is assumed to employ a high-gain directional antenna, with a gain on the order of 40dBi. This assumption is consistent with reported gains for modern Ku-band satellite terminals operating at similar frequencies [14]. Beam visibility is determined by satellite position, beam footprint geometry, and elevation angle constraints.

Beam congestion is modeled using a population density map to capture spatial variations in traffic demand across the satellite footprint. To account for temporal fluctuations in network usage, the effective load of each beam is further modulated by time-of-day effects. This modeling approach provides a simplified yet realistic abstraction of congestion dynamics in LEO satellite networks, where traffic demand varies across both geography and time. The normalized load of beam b at time t is defined in Equation 1.

$$\text{Load}_b(t) = \text{clip}_{[0,1]}(L_{\min,b} + (L_{\max,b} - L_{\min,b})\phi_b(t) + \varepsilon_b) \quad (1)$$

where $\text{clip}_{[0,1]}(\cdot)$ constrains the load to the valid interval $[0, 1]$.

The parameters $L_{\min,b}$ and $L_{\max,b}$ represent the minimum and maximum load levels for beam b , respectively, and are defined as functions of the normalized population density ρ_b within the beam footprint, as shown in Equation 2. The calibration parameters enforce a non-zero traffic floor under low demand and allow load saturation in highly populated regions.

$$L_{\min,b} = \beta_{\min}(\rho_b), \quad L_{\max,b} = \beta_{\max}(\rho_b), \quad (2)$$

Temporal variation in traffic demand is modeled using a periodic function, as shown in Equation 3. T denotes the period of the cycle and ψ_b is a beam-specific phase offset. The phase shift ψ_b is introduced to prevent all beams from reaching peak load simultaneously, reflecting the fact that peak traffic periods occur at different local times across geographic regions and time zones. Finally, $\varepsilon_b(t)$ denotes a stochastic noise term that captures short-term traffic fluctuations and uncertainty in beam load estimation.

$$\phi_b(t) = 0.5 + 0.5 \sin\left(2\pi\left(\frac{t}{T} + \psi_b\right) - \frac{\pi}{2}\right), \quad (3)$$

The aircraft carries a variable number of passengers, each generating traffic according to an application usage model. The application type for each passenger is sampled from a predefined distribution. The aggregate traffic demand and latency sensitivity of the aircraft therefore vary dynamically based on passenger count and application mix.

To mitigate frequent or oscillatory handovers, we impose a dwell time constraint, requiring the UT to remain connected to the serving beam for a minimum duration following a handover. In addition, we employ a time-to-trigger (TTT) mechanism, whereby a candidate beam must consistently remain the most favorable option for a predefined interval before a handover can be initiated.

IV. PROPOSED TECHNIQUE

A. Proposed Methodology

The proposed framework operates in a closed-loop manner along the aircraft trajectory. At each decision step, the aircraft position is updated and all visible candidate beams are identified based on link geometry and visibility constraints. The handover management agent then constructs an observation from current link conditions, estimated beam load, quality-of-experience indicators, and mobility control timers. Using this observation, the agent selects a beam. To ensure stability, dwell time and time-to-trigger constraints are applied before executing any handover, preventing premature or oscillatory switching. Once a handover decision is finalized, the environment updates the serving beam and computes the resulting quality-of-experience metrics and reward, which are used to iteratively refine the agent's policy throughout the route. This process is described in Figure 1.

We propose an ODT agent for handover management. Decision Transformers formulate reinforcement learning as a conditional sequence modeling problem, analogous to language modeling, in which the policy predicts the next action conditioned on past states, past actions, and a desired return-to-go [15]. Transformer-based decision architectures have recently been explored in networking contexts for intent-driven network management and hierarchical resource optimization, demonstrating their potential for adaptive and goal-aware control in complex communication systems [16].

Online Decision Transformers extend this framework to online reinforcement learning, enabling continuous policy improvement through direct interaction with the environment. By combining sequence modeling with policy gradient-based optimization, ODTs effectively bridge offline pretraining and online adaptation. The agent maintains an experience buffer of trajectories, where each trajectory consists of state sequences, action sequences, and corresponding returns-to-go, which are used to incrementally update the policy during training [8].

To train the ODT, an initial offline dataset of handover trajectories is synthetically generated using the proposed simulation environment. The dataset is constructed by executing multiple simulation episodes under varying handover strategies, thereby producing diverse sequences of observations, actions, and achieved rewards. These trajectories capture a wide range of handover behaviors and performance trade-offs and are used to initialize the ODT prior to online fine-tuning through continued interaction with the environment.

For each trajectory, the return-to-go is computed as the discounted cumulative sum of future rewards using a discount factor of γ . The return-to-go at time step t is defined as the sum of rewards from that time onward, discounted over the remaining horizon. During offline dataset construction, returns-to-go are computed via a backward pass over each trajectory segment. During online training, the same discounted accumulation is applied to rewards stored in the experience buffer when forming training batches, as shown in algorithm 1. This return computation enables the ODT to condition its

action predictions on long-term performance objectives rather than instantaneous rewards.

Algorithm 1 ODT Inference for LEO Handover

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1: Input: route  $\mathcal{R}$ , trained ODT policy  $\pi_\theta$ , context length  $K$ ,
   dwell time  $T_{\text{dwell}}$ , time-to-trigger  $T_{\text{ttt}}$ , target return  $\hat{R}_0$ 
2: Init: serving beam  $b$ , timers, context deque  $(s, a, r) \leftarrow []$ 
3: for each step  $t$  along  $\mathcal{R}$  do
4:   Update aircraft position and candidate beams
5:   Build observation  $o_t$  and valid action mask  $\mathcal{M}_t$ 
6:   if no valid actions in  $\mathcal{M}_t$  then
7:     Set  $a_t \leftarrow -1$  (penalty)
8:     continue
9:   end if
10:  Construct ODT input sequence from recent context
11:  Compute masked action logits from  $\pi_\theta$ 
12:  Select action:
      
$$\tilde{a}_t \leftarrow \arg \max_{a \in \mathcal{M}_t} \text{logit}(a)$$

13:  if dwell/TTT is satisfied then
14:    map  $\tilde{a}_t \rightarrow a_t$ 
15:  else
16:    keep  $b$ 
17:  end if
18:  Compute QoE metrics and reward  $r_t$ 
19:  Update context deque with  $(o_t, a_t, r_t)$ 
20: end for

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The reward function, shown in Eq.(4), is formulated as a weighted combination of bandwidth allocation efficiency, α , and latency satisfaction, L , defined in Eqs.(5) and (6), respectively. A penalty, D , is applied when an action results in a service interruption.

$$R = w_{\text{alloc}} \cdot \alpha + w_{\text{lat}} \cdot L - \lambda_{\text{drop}} \cdot D \quad (4)$$

$$\alpha = \frac{BW_{\text{allocated}}}{BW_{\text{demanded}}} \quad (5)$$

$$L = \exp\left(-\frac{\max(0, L - L_{\text{req}})}{L_{\text{req}}}\right) \quad (6)$$

B. Simulation Environment

The proposed system model and handover framework are implemented and evaluated in a Python-based simulation environment. To simulate realistic satellite dynamics and network behavior, the environment is adapted from the open-source multi-agent LEO packet routing simulator proposed in [9]. While the original framework focuses on packet routing, it is extended in this work to model handover decision-making by incorporating spot beam structures, an airborne user terminal, passenger-level traffic demand and application-specific performance requirements.

The simulator takes a real or synthetic aircraft trajectory as input, defined by longitude, latitude, altitude, velocity, and time. Simulation time progresses according to aircraft motion,

TABLE I
APPLICATION-LEVEL SERVICE REQUIREMENTS

Application	Data Rate Demand	Latency Requirement
Streaming	25 Mbps	300 ms
VoIP	0.1 Mbps	100 ms
Email	0.5 Mbps	200 ms
Web browsing	1 Mbps	500 ms
File transfer	10 Mbps	1000 ms
Video conferencing	2 Mbps	50 ms

enabling realistic evaluation of handover behavior along full flight paths.

Table I summarizes the representative application-level data rate and latency requirements used in the simulation, derived from 3GPP service requirement specifications [17] and traffic characterization reported in the Cisco Visual Networking Index [18].

To assess the robustness and adaptability of the proposed handover strategy, we simulate the following scenarios:

- **Varying Load Cycle Scenarios:** Beam load is modeled as a function of population density and time-of-day effects. The number of congestion cycles per episode is varied to evaluate the agent's ability to adapt to changing network conditions and recover from periods of high congestion.
- **Large Aircraft Scenario:** The number of passengers is increased from 5 to 15, resulting in higher aggregate traffic demand and more stringent resource constraints.

We test each scenario over multiple independent environment seeds in which passenger demand and load realizations are resampled. The reported results correspond to the mean performance across these test episodes.

V. PERFORMANCE EVALUATION

The performance of the proposed ODT agent is evaluated against established RL baselines, including DQN and PPO, as well as an SNR-based greedy strategy serving as a non-learning benchmark. We conduct the evaluation across the scenarios described in Section III, including varying load cycles and aircraft sizes. Performance is assessed using key metrics such as allocation-to-demand ratio, which quantifies the fraction of requested throughput successfully delivered, along with service interruption duration and latency. Collectively, these metrics capture complementary dimensions of handover effectiveness, including demand satisfaction, responsiveness, and service continuity.

Table II summarizes the performance of each agent across all evaluation scenarios. Across all cases, the average allocation-to-demand ratios are high and largely comparable across agents, with values exceeding 98% in most scenarios. However, allocation-to-demand begins to degrade in the large aircraft scenario, where the number of users and aggregate

TABLE II
SUMMARY OF PERFORMANCE METRICS ACROSS AGENTS

Scenario	Allocation to Demand (%)				Latency (ms)				Service Interruption (s)			
	ODT	DQN	PPO	BASE	ODT	DQN	PPO	BASE	ODT	DQN	PPO	BASE
Load Cycle 1	99.62	99.09	99.86	98.5	56.6	56.29	56.08	56.81	6.54	25.97	0.50	51.88
Load Cycle 2	99.49	99.5	99.97	99.4	56.19	55.87	55.46	56.08	4.99	6.58	5.56	10.98
Load Cycle 3	99.45	99.1	99.34	98.9	56.02	55.84	56.16	55.97	11.95	24.33	15.92	32.2
Large Aircraft	73.08	73.5	74.34	73.1	270.52	292.04	300.32	293.76	6.54	23.49	0.50	51.81

traffic demand increase significantly, resulting in a reduction to approximately 73%, reflecting the impact of capacity constraints under higher load conditions. Similarly, latency remains around 55 ms across all agents and most scenarios, but increases substantially in the large aircraft case, likely due to elevated queuing delays resulting from unmet demand and sustained congestion.

Service interruption is defined as the total duration, over an entire episode, during which user demand is completely unmet. We observe that both PPO and ODT consistently outperform the SNR-based greedy baseline and DQN in terms of service interruption across all scenarios. In Load Cycle 1, PPO achieves the lowest interruption time at 0.50 s, followed by ODT at 6.54 s, representing a substantial reduction compared to DQN at 25.97 s and the baseline at 51.88 s. In Load Cycle 2, ODT achieves the lowest interruption at 4.99 s, narrowly surpassing PPO at 5.56 s and clearly improving upon DQN at 6.58 s and the baseline at 10.98 s. In Load Cycle 3, ODT further distinguishes itself with 11.95 s of interruption, compared to 15.92 s for PPO, while both remain significantly better than DQN at 24.33 s and the baseline at 32.2 s. Under sustained high demand in the Large Aircraft scenario, PPO again achieves the lowest interruption at 0.50 s, with ODT at 6.54 s still markedly outperforming DQN at 23.49 s and the baseline at 51.81 s.

A direct comparison between PPO and ODT reveals distinct behavioral characteristics. PPO demonstrates strong reactivity, achieving minimal interruption under short-lived congestion spikes, as observed in Load Cycle 1 and the Large Aircraft scenario. In contrast, ODT exhibits more consistent performance as congestion patterns become structured and cyclic, particularly in Load Cycles 2 and 3. This indicates that leveraging longer decision histories can be beneficial when network load evolves gradually rather than through isolated bursts. Such recurring demand variations are common in satellite communication systems, especially in aeronautical networks where traffic fluctuates with flight density, time of day, and seasonal travel trends. In our prior work, we examined these traffic variations and their impact on congestion in satellite networks [19], further underscoring the value of incorporating longer-term performance trends into handover decision-making.

Figure 2 illustrates the allocated versus requested throughput under Load Cycle 3. We observe that ODT closely tracks

Allocated vs Requested Throughput (Load Cycle 3)

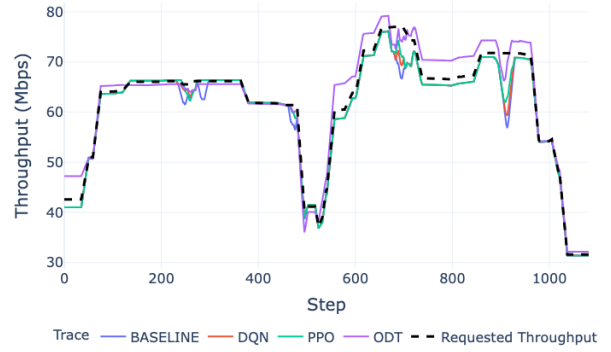


Fig. 2. Comparison of allocated and requested throughput across agents under the Load Cycle 3 scenario.

Throughput Delta(load_cycle_5)

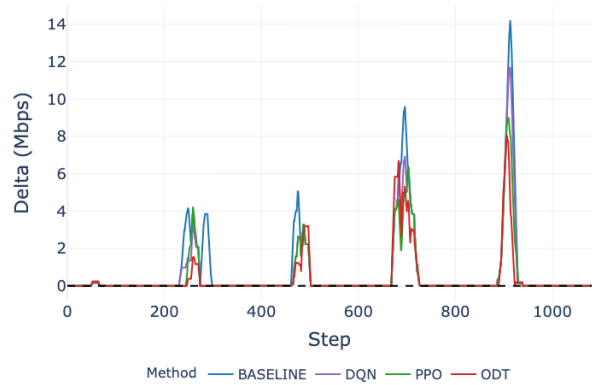


Fig. 3. Comparison of the difference between requested and allocated throughput across agents for the Load Cycle 3 scenario

the requested throughput during cyclic demand variations. Particularly, the ODT agent exhibits a faster recovery from the congestion that occurs between steps 800 and 1000. This behavior is further reflected in Figure 3, which displays the deltas between the requested and allocated throughput across all agents. The ODT agent demonstrates rapid recovery with relatively contained delta spikes compared to the baseline and

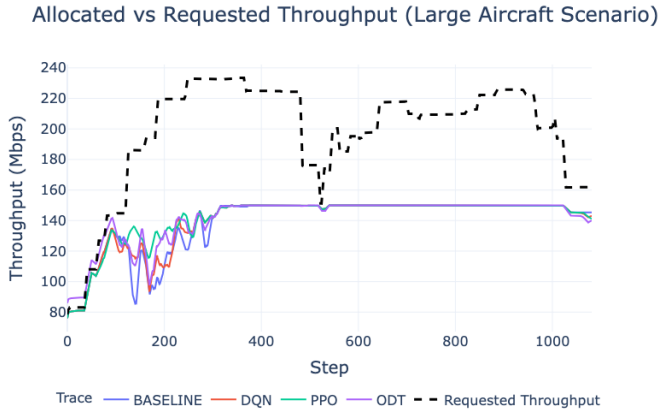


Fig. 4. Comparison of allocated and requested throughput across agents under the large aircraft scenario

DQN. Although PPO also responds quickly to sudden changes, ODT exhibits smoother correction behavior across repeated cyclic transitions, suggesting that conditioning on extended trajectory context helps anticipate recurring congestion patterns rather than reacting purely to instantaneous load changes.

In the Large Aircraft scenario, the allocated throughput for all agents saturates at approximately 145 Mbps while requested throughput exceeds 200 Mbps for extended intervals, as shown in Figure 4. Although PPO achieves minimal service interruption in this setting, the allocation-to-demand ratio remains approximately 73%, indicating that a substantial portion of requested bandwidth is not delivered. From a user experience perspective, this reflects sustained congestion and degraded service quality, regardless of interruption minimization.

In contrast, ODT achieves lower latency under this persistent high-demand condition, suggesting improved coordination of handover decisions even when total capacity is insufficient to satisfy aggregate demand. The plots therefore highlight that minimizing interruption alone does not guarantee satisfactory user experience when system-level capacity constraints dominate performance.

VI. CONCLUSION

In this paper, we proposed an ODT-based approach for handover management in LEO satellite networks, addressing challenges such as high satellite mobility, dynamic congestion, and varying user demand. By framing handover as a sequence modeling problem, the proposed method allows decisions to consider longer-term effects, including delayed congestion and handover costs that are difficult to capture with conventional reinforcement learning or heuristic strategies.

Simulation results show that ODT delivers stable and competitive performance across diverse traffic conditions. Allocation-to-demand remains similar across agents under moderate load, but differences emerge under structured congestion patterns. In cyclic load scenarios, ODT demonstrates

more stable service interruption behavior, while under sustained high demand it achieves lower latency compared to other methods. These findings suggest that incorporating longer temporal context can be beneficial when congestion evolves over time rather than occurring as isolated spikes.

It is important to note that ODT relies on the quality and diversity of its training data. In this work, the offline dataset was synthetically generated using the proposed simulation environment to model representative mobility and load dynamics. While this enables controlled evaluation, future work may incorporate real operational data to further improve generalization and responsiveness to complex traffic behavior. Leveraging real traffic traces could enhance short-term reactivity while preserving the longer-term awareness observed in this study.

In our prior work, we introduced a transformer-based model for preflight QoS prediction in satellite communication networks [20]. Integrating predictive QoS estimation with multi-orbit handover optimization represents a promising next step. If performance degradation can be anticipated in advance along a flight segment, the handover policy could proactively select an alternative orbit or frequency band to maintain service continuity.

Future extensions will also consider multi-band and multi-orbit satellite systems, where aircraft terminals may access heterogeneous networks spanning different frequency bands and constellations, including both LEO and GEO systems. A sequence-based approach such as ODT is naturally suited to this setting, as it can incorporate historical and predicted performance trends when selecting among links with different latency and congestion characteristics.

Overall, these results highlight the potential of transformer-based decision models for more adaptive and temporally aware handover optimization in next-generation satellite communication systems.

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